

Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

L-3/T-2

July 2025

CSE 311: Data Communication

Time: 25 minutes

Marks: 20

Student Name: _____

Student ID: _____

1. Consider, you are sending some binary data from one device to another. (10)
 - (a) Suppose, you have data elements 01011011. Now, convert them to signal elements according to the MLT-3. (5)
Use vertical dashed bar to differentiate each bit while drawing.
 - (b) “If a scheme exhibits baseline wandering, it will always exhibits DC elements.” Do you agree with the statement? Justify your answer. If you disagree with the statement, then mention a scheme as a counter example. (5)

2. Consider, you are transmitting signals using pulse modulation. (10)
- (a) Suppose, you are using PWM as your modulation technique. Now, draw the sawtooth signal if you want to modulate - (5)
- i. Leading Edge
 - ii. Trailing Edge
 - iii. Both the Leading edge and Trailing Edge
 - iv. Leading Edge and Trailing Edge repetitively (i.e., first pulse uses leading edge modulation, second pulse uses trailing edge modulation, and so on repetitively)
 - v. Leading Edge, Both Edge and Trailing Edge repetitively (i.e., first pulse uses leading edge modulation, second pulse uses both edge modulation, third pulse uses trailing edge modulation and so on repetitively)
- (b) Now, you are receiving signals which are modulated using PWM. How can you detect the signals? (5)
- Draw a flowchart of the detection.